MAREABIO_REL — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Nota sobre este algoritmo

Tres fases biologicas con sigma_rel=True. Gana a MAREA_REL a 5k FES y pierde a 50k: el ciclo biologico es arranque rapido, no convergencia.

Resumen — posicion entre los 11 algoritmos

dimension	presupuesto	rango medio de Friedman	posicion de 11	funciones donde es el mejor
D10	5k FES	5.20	5º	1 de 71
D10	50k FES	6.21	7º	0 de 71
D10	500k FES	7.20	8º	1 de 71
D50	5k FES	4.65	4º	3 de 71
D50	50k FES	4.73	5º	0 de 71
D50	500k FES	7.17	9º	0 de 71

MAREABIO_REL — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	1.04e+05	2.86e+07	5.82e+06	7.37e+06	6.08e+06	3.94e+04	7.14e+05	1.78e+05	2.39e+05	1.88e+05	8.54e+03	7.81e+04	2.85e+04	3.14e+04	1.46e+04
F22014	2014	10	2.26e+06	9.04e+06	5.41e+06	5.55e+06	1.57e+06	9.86e+05	3.99e+06	2.64e+06	2.59e+06	6.82e+05	7.86e+05	2.51e+06	1.67e+06	1.62e+06	3.81e+05
F32014	2014	10	3.54e+03	5.60e+04	3.00e+04	3.11e+04	1.47e+04	2.96e+01	9.73e+02	1.43e+02	2.62e+02	2.47e+02	5.19e+00	2.09e+01	1.39e+01	1.34e+01	3.70e+00
F42014	2014	10	7.22e-01	1.02e+02	3.69e+01	3.88e+01	2.28e+01	4.69e-01	6.85e+01	3.49e+01	2.80e+01	1.63e+01	1.64e-01	3.49e+01	3.48e+01	2.66e+01	1.43e+01
F52014	2014	10	5.27e+00	2.07e+01	2.06e+01	2.03e+01	2.15e+00	4.73e+00	2.05e+01	2.04e+01	2.01e+01	2.20e+00	3.98e+00	2.04e+01	2.03e+01	2.00e+01	2.28e+00
F62014	2014	10	2.81e+00	9.43e+00	6.08e+00	6.06e+00	1.52e+00	1.70e+00	8.52e+00	5.33e+00	5.10e+00	1.49e+00	8.97e-01	8.38e+00	4.33e+00	4.53e+00	1.53e+00
F72014	2014	10	8.07e-01	1.12e+00	1.03e+00	1.01e+00	7.60e-02	6.24e-01	1.01e+00	8.60e-01	8.56e-01	7.70e-02	3.99e-01	9.25e-01	6.91e-01	6.94e-01	9.90e-02
F82014	2014	10	1.65e+01	8.49e+01	3.51e+01	3.60e+01	1.27e+01	1.58e+01	5.70e+01	3.08e+01	3.24e+01	9.64e+00	1.49e+01	4.96e+01	2.83e+01	2.81e+01	8.00e+00
F92014	2014	10	1.69e+01	8.02e+01	3.64e+01	3.73e+01	1.26e+01	1.48e+01	7.47e+01	3.43e+01	3.45e+01	1.23e+01	1.30e+01	6.38e+01	2.85e+01	3.01e+01	1.05e+01
F102014	2014	10	4.09e+02	1.57e+03	1.02e+03	1.01e+03	3.24e+02	2.02e+02	1.49e+03	7.60e+02	8.09e+02	3.13e+02	1.66e+02	1.36e+03	5.10e+02	5.29e+02	2.18e+02
F112014	2014	10	5.10e+02	1.92e+03	1.13e+03	1.13e+03	3.31e+02	3.14e+02	1.88e+03	9.08e+02	9.16e+02	3.14e+02	2.64e+02	1.31e+03	7.82e+02	7.71e+02	2.63e+02
F122014	2014	10	7.35e-01	2.50e+00	1.52e+00	1.52e+00	3.74e-01	2.72e-01	1.45e+00	9.59e-01	9.45e-01	2.38e-01	2.72e-01	1.05e+00	6.31e-01	6.41e-01	1.30e-01
F132014	2014	10	2.06e-01	1.16e+00	4.62e-01	5.12e-01	2.12e-01	8.34e-02	7.72e-01	3.15e-01	3.42e-01	1.39e-01	6.19e-02	3.12e-01	1.50e-01	1.61e-01	5.73e-02
F142014	2014	10	1.52e-01	9.53e-01	3.58e-01	4.33e-01	2.00e-01	7.42e-02	4.30e-01	2.48e-01	2.35e-01	8.70e-02	1.53e-02	1.26e-01	5.15e-02	5.54e-02	2.34e-02
F152014	2014	10	2.76e+00	1.11e+01	5.36e+00	5.47e+00	1.66e+00	1.45e+00	5.23e+00	3.06e+00	3.09e+00	8.32e-01	7.73e-01	3.33e+00	1.94e+00	2.03e+00	4.94e-01
F162014	2014	10	2.61e+00	4.05e+00	3.42e+00	3.41e+00	3.47e-01	1.71e+00	3.61e+00	2.89e+00	2.88e+00	3.66e-01	9.83e-01	3.11e+00	2.50e+00	2.47e+00	4.19e-01
F172014	2014	10	9.67e+03	5.99e+05	3.16e+04	7.03e+04	1.03e+05	3.36e+03	2.75e+04	1.06e+04	1.17e+04	5.50e+03	6.17e+02	3.52e+03	1.70e+03	1.77e+03	6.69e+02
F182014	2014	10	9.99e+03	1.00e+05	3.87e+04	4.08e+04	1.74e+04	6.77e+03	8.98e+04	3.61e+04	3.75e+04	1.59e+04	5.79e+03	5.74e+04	2.28e+04	2.42e+04	1.05e+04
F192014	2014	10	7.34e+00	5.71e+03	1.83e+01	1.37e+02	7.96e+02	5.50e+00	1.23e+02	1.45e+01	1.94e+01	2.05e+01	3.86e+00	1.03e+02	1.15e+01	1.35e+01	1.38e+01
F202014	2014	10	2.42e+03	5.65e+04	2.30e+04	2.42e+04	1.39e+04	1.17e+03	3.98e+04	1.51e+04	1.66e+04	9.27e+03	4.75e+01	1.58e+03	5.21e+02	5.55e+02	3.42e+02
F212014	2014	10	2.63e+03	1.95e+05	1.86e+04	2.72e+04	3.09e+04	1.95e+03	5.12e+04	1.18e+04	1.38e+04	9.46e+03	5.45e+02	2.24e+04	4.26e+03	5.88e+03	5.46e+03
F222014	2014	10	5.77e+01	1.68e+03	3.69e+02	4.81e+02	3.25e+02	2.35e+01	6.53e+02	2.43e+02	1.71e+02	1.26e+02	2.35e+01	6.49e+02	2.41e+02	1.69e+02	1.26e+02
F232014	2014	10	2.37e+02	2.55e+02	2.42e+02	2.44e+02	3.51e+00	2.37e+02	2.42e+02	2.42e+02	2.42e+02	1.74e+00	2.37e+02	2.42e+02	2.42e+02	2.42e+02	1.75e+00
F242014	2014	10	1.24e+02	2.13e+02	1.55e+02	1.59e+02	2.24e+01	1.24e+02	2.07e+02	1.54e+02	1.57e+02	2.06e+01	1.23e+02	2.06e+02	1.48e+02	1.50e+02	1.76e+01
F252014	2014	10	2.02e+02	2.13e+02	2.05e+02	2.06e+02	2.59e+00	1.98e+02	2.13e+02	2.05e+02	2.04e+02	3.18e+00	1.84e+02	2.09e+02	2.03e+02	2.03e+02	3.82e+00
F262014	2014	10	1.00e+02	1.01e+02	1.00e+02	1.00e+02	2.38e-01	1.00e+02	1.01e+02	1.00e+02	1.00e+02	1.95e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	9.50e-02
F272014	2014	10	3.38e+02	6.01e+02	4.06e+02	4.35e+02	5.75e+01	3.36e+02	5.99e+02	4.01e+02	4.25e+02	5.27e+01	3.28e+02	5.39e+02	4.00e+02	4.19e+02	4.35e+01
F282014	2014	10	4.46e+02	1.30e+03	7.37e+02	7.52e+02	1.83e+02	4.15e+02	1.17e+03	6.31e+02	6.46e+02	1.57e+02	1.62e+02	1.16e+03	5.44e+02	5.65e+02	1.73e+02
F292014	2014	10	5.06e+06	4.70e+07	1.52e+07	1.68e+07	8.50e+06	4.50e+06	1.91e+07	6.81e+06	7.74e+06	3.54e+06	4.50e+06	1.90e+07	5.80e+06	7.38e+06	3.14e+06
F302014	2014	10	1.09e+04	1.55e+08	3.68e+05	9.46e+06	2.73e+07	2.16e+03	1.01e+07	8.60e+03	4.34e+05	1.98e+06	1.90e+03	1.01e+07	5.39e+03	4.24e+05	1.98e+06
F12017	2017	10	2.87e+06	9.13e+06	6.31e+06	6.26e+06	1.57e+06	1.53e+06	4.62e+06	2.81e+06	2.84e+06	6.18e+05	9.37e+05	2.53e+06	1.69e+06	1.74e+06	3.59e+05
F22017	2017	10	5.96e+00	4.27e+03	2.07e+02	5.02e+02	8.24e+02	1.96e+00	7.48e+00	4.89e+00	4.76e+00	1.25e+00	1.24e+00	4.35e+00	2.63e+00	2.63e+00	6.41e-01
F32017	2017	10	1.62e+00	1.25e+02	1.29e+01	3.33e+01	3.60e+01	4.81e-01	6.85e+01	5.15e+00	8.18e+00	1.25e+01	3.91e-01	6.80e+01	4.80e+00	6.78e+00	1.26e+01
F42017	2017	10	3.98e+01	9.53e+01	7.07e+01	6.98e+01	1.01e+01	3.89e+01	6.37e+01	5.04e+01	5.10e+01	6.21e+00	2.45e+01	4.77e+01	3.95e+01	3.87e+01	5.09e+00
F52017	2017	10	1.03e-06	5.12e-03	2.57e-05	3.09e-04	9.53e-04	5.36e-07	3.98e-04	3.75e-06	2.78e-05	7.20e-05	1.01e-07	3.19e-04	7.18e-07	1.59e-05	5.71e-05
F62017	2017	10	1.54e+02	3.97e+02	2.87e+02	2.78e+02	4.89e+01	1.38e+02	2.32e+02	1.92e+02	1.89e+02	1.99e+01	1.03e+02	1.79e+02	1.52e+02	1.50e+02	1.55e+01
F72017	2017	10	3.00e+01	1.66e+02	6.40e+01	6.89e+01	2.90e+01	1.80e+01	8.20e+01	3.80e+01	4.05e+01	1.59e+01	1.21e+01	3.80e+01	1.80e+01	1.87e+01	5.60e+00
F82017	2017	10	6.88e-03	5.59e+00	7.36e-01	1.38e+00	1.58e+00	3.81e-03	3.55e+00	1.80e-01	5.49e-01	7.77e-01	1.77e-03	1.82e+00	4.41e-03	1.45e-01	2.99e-01
F92017	2017	10	3.94e+02	1.77e+03	1.10e+03	1.07e+03	3.28e+02	1.96e+02	1.51e+03	9.94e+02	9.14e+02	2.95e+02	1.77e+02	1.32e+03	7.26e+02	6.72e+02	2.55e+02
F102017	2017	10	2.79e+03	6.28e+04	1.67e+04	2.11e+04	1.22e+04	6.00e+01	2.67e+04	8.00e+03	9.22e+03	7.57e+03	3.17e+01	3.64e+03	2.03e+02	6.01e+02	7.73e+02
F112017	2017	10	3.49e+05	4.11e+07	4.48e+06	8.03e+06	8.83e+06	6.08e+04	2.63e+06	5.04e+05	6.53e+05	5.65e+05	2.98e+04	1.61e+05	6.95e+04	7.13e+04	2.86e+04
F122017	2017	10	7.54e+03	6.89e+05	3.47e+04	9.82e+04	1.50e+05	4.95e+03	1.44e+05	1.76e+04	2.63e+04	2.92e+04	2.91e+03	2.47e+04	1.04e+04	1.14e+04	5.64e+03
F132017	2017	10	6.98e+03	6.22e+04	2.22e+04	2.34e+04	1.10e+04	5.51e+02	1.97e+04	5.86e+03	6.44e+03	3.99e+03	4.03e+01	9.79e+01	6.33e+01	6.48e+01	1.46e+01
F142017	2017	10	5.94e+03	7.02e+05	4.30e+04	1.37e+05	1.68e+05	2.57e+03	3.84e+04	1.34e+04	1.37e+04	8.58e+03	5.32e+01	1.04e+04	2.27e+02	7.33e+02	1.95e+03
F152017	2017	10	3.42e+02	3.57e+04	5.65e+03	9.37e+03	9.90e+03	1.10e+01	1.28e+04	5.29e+02	1.25e+03	2.43e+03	3.86e+00	7.58e+02	2.39e+02	2.62e+02	2.43e+02
F162017	2017	10	1.99e+02	3.75e+04	2.32e+03	4.28e+03	6.49e+03	4.30e+01	7.65e+02	7.04e+01	2.08e+02	2.40e+02	2.90e+01	6.63e+02	3.98e+01	1.16e+02	1.72e+02
F172017	2017	10	6.95e+03	6.49e+04	2.48e+04	2.66e+04	1.23e+04	4.20e+03	5.89e+04	2.19e+04	2.35e+04	1.15e+04	1.85e+03	3.31e+04	1.20e+04	1.38e+04	6.86e+03
F182017	2017	10	1.22e+03	6.82e+04	1.06e+04	1.46e+04	1.29e+04	3.85e+01	1.67e+04	2.92e+03	4.18e+03	3.77e+03	1.15e+01	9.53e+01	3.53e+01	3.82e+01	1.63e+01
F192017	2017	10	4.21e+01	4.50e+02	1.24e+02	1.48e+02	9.56e+01	3.39e+01	2.83e+02	5.17e+01	8.71e+01	7.82e+01	2.60e+01	2.51e+02	3.39e+01	6.72e+01	6.96e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.23e+02	3.08e+02	1.42e+02	1.58e+02	4.55e+01	1.16e+02	2.49e+02	1.28e+02	1.41e+02	3.79e+01	1.16e+02	2.39e+02	1.23e+02	1.35e+02	3.72e+01
F212017	2017	10	1.10e+02	1.37e+02	1.20e+02	1.20e+02	6.18e+00	1.08e+02	1.35e+02	1.16e+02	1.17e+02	5.85e+00	1.06e+02	1.22e+02	1.13e+02	1.13e+02	3.67e+00
F222017	2017	10	1.62e+02	2.84e+02	2.41e+02	2.25e+02	3.33e+01	1.60e+02	2.41e+02	2.26e+02	2.06e+02	3.23e+01	1.53e+02	2.33e+02	2.24e+02	1.96e+02	3.32e+01
F232017	2017	10	2.06e+02	3.73e+02	3.21e+02	3.17e+02	3.40e+01	1.71e+02	3.52e+02	2.89e+02	2.85e+02	3.34e+01	1.55e+02	3.38e+02	2.69e+02	2.67e+02	3.40e+01
F242017	2017	10	4.82e+02	6.70e+02	5.09e+02	5.22e+02	4.40e+01	4.79e+02	6.13e+02	4.84e+02	4.99e+02	2.96e+01	4.78e+02	5.83e+02	4.80e+02	4.96e+02	2.51e+01
F252017	2017	10	2.56e+00	6.89e+02	5.26e+02	4.82e+02	1.94e+02	1.72e+00	6.87e+02	4.89e+02	4.50e+02	1.87e+02	1.58e+00	6.35e+02	4.89e+02	4.44e+02	1.83e+02
F262017	2017	10	4.05e+02	6.06e+02	4.60e+02	4.68e+02	4.94e+01	3.99e+02	5.72e+02	4.26e+02	4.46e+02	4.53e+01	3.90e+02	5.72e+02	4.10e+02	4.25e+02	3.75e+01
F272017	2017	10	4.71e+01	6.14e+02	1.90e+02	2.57e+02	1.98e+02	3.69e+01	5.00e+01	4.42e+01	4.39e+01	3.49e+00	3.02e+01	4.08e+01	3.81e+01	3.77e+01	2.27e+00
F282017	2017	10	1.44e+03	3.95e+06	3.20e+04	2.67e+05	8.24e+05	1.08e+03	3.26e+04	1.44e+03	4.18e+03	6.30e+03	1.06e+03	3.25e+04	1.37e+03	3.37e+03	5.69e+03
F292017	2017	10	3.57e+04	5.00e+08	3.12e+05	1.79e+07	7.38e+07	1.01e+04	6.46e+06	3.15e+04	1.64e+05	9.00e+05	6.19e+03	4.57e+04	2.03e+04	2.15e+04	9.16e+03
F12022	2022	10	5.32e+00	6.67e+02	1.11e+01	6.03e+01	1.32e+02	1.94e+00	5.78e+00	3.73e+00	3.75e+00	8.91e-01	6.96e-01	3.28e+00	2.26e+00	2.33e+00	5.68e-01
F22022	2022	10	6.40e-01	9.19e+01	2.31e+01	3.48e+01	3.20e+01	1.85e-01	7.66e+01	4.72e+00	1.48e+01	2.48e+01	1.52e-01	7.59e+01	4.38e+00	1.19e+01	2.24e+01
F32022	2022	10	1.22e-04	4.86e-04	3.09e-04	3.03e-04	7.20e-05	6.05e-05	2.21e-04	1.55e-04	1.53e-04	3.94e-05	4.86e-05	1.35e-04	1.01e-04	9.88e-05	2.21e-05
F42022	2022	10	2.69e+01	1.40e+02	6.00e+01	5.94e+01	2.27e+01	2.00e+01	8.82e+01	4.00e+01	4.25e+01	1.54e+01	1.20e+01	5.80e+01	2.00e+01	2.14e+01	7.54e+00
F52022	2022	10	8.24e-03	5.45e+00	1.10e+00	1.61e+00	1.41e+00	2.09e-03	4.81e+00	9.14e-01	1.10e+00	9.99e-01	1.22e-03	1.37e+00	4.57e-01	4.42e-01	3.55e-01
F62022	2022	10	2.49e+04	1.70e+05	7.38e+04	7.78e+04	3.20e+04	1.65e+04	8.72e+04	3.88e+04	4.31e+04	1.59e+04	9.23e+03	8.49e+04	3.40e+04	3.60e+04	1.53e+04
F72022	2022	10	2.36e+01	1.93e+02	5.40e+01	6.90e+01	3.93e+01	1.69e+01	1.47e+02	3.58e+01	4.58e+01	3.17e+01	1.14e+01	1.40e+02	2.93e+01	3.92e+01	3.21e+01
F82022	2022	10	3.08e+01	8.86e+02	7.31e+01	1.65e+02	1.99e+02	1.53e+01	3.33e+01	2.79e+01	2.77e+01	2.29e+00	9.29e+00	2.74e+01	2.56e+01	2.52e+01	2.43e+00
F92022	2022	10	2.99e+00	7.12e+02	4.09e+02	3.05e+02	1.99e+02	1.40e+00	7.09e+02	3.59e+02	2.49e+02	1.83e+02	9.79e-01	7.09e+02	3.59e+02	2.49e+02	1.83e+02
F102022	2022	10	2.22e+02	1.35e+03	2.29e+02	3.07e+02	1.66e+02	2.21e+02	4.57e+02	2.27e+02	2.80e+02	6.95e+01	2.20e+02	4.45e+02	2.23e+02	2.55e+02	5.76e+01
F112022	2022	10	9.60e-01	1.04e+03	1.47e+00	1.27e+02	3.01e+02	5.14e-01	9.29e+02	9.10e-01	7.20e+01	2.46e+02	4.00e-01	9.29e+02	7.03e-01	7.10e+01	2.44e+02
F122022	2022	10	1.70e+02	3.36e+02	2.21e+02	2.29e+02	4.63e+01	1.67e+02	3.25e+02	1.91e+02	2.00e+02	3.75e+01	1.67e+02	2.26e+02	1.74e+02	1.81e+02	1.57e+01

MAREABIO_REL — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	1.59e+08	1.14e+09	4.60e+08	5.12e+08	2.29e+08	1.26e+07	3.37e+07	2.46e+07	2.42e+07	5.63e+06	6.15e+06	2.08e+07	1.21e+07	1.24e+07	3.24e+06
F22014	2014	50	8.04e+09	4.14e+10	1.83e+10	1.93e+10	6.34e+09	2.07e+08	3.30e+08	2.53e+08	2.56e+08	2.91e+07	1.34e+08	2.21e+08	1.88e+08	1.85e+08	1.95e+07
F32014	2014	50	1.37e+05	3.88e+05	2.30e+05	2.35e+05	4.78e+04	8.54e+03	4.98e+04	2.32e+04	2.38e+04	9.25e+03	4.14e+02	8.40e+02	5.66e+02	5.71e+02	8.98e+01
F42014	2014	50	3.14e+03	1.23e+04	7.21e+03	7.40e+03	2.43e+03	5.60e+01	3.25e+02	1.83e+02	1.86e+02	5.82e+01	4.58e+01	2.47e+02	1.46e+02	1.41e+02	4.74e+01
F52014	2014	50	2.11e+01	2.13e+01	2.13e+01	2.13e+01	4.64e-02	2.11e+01	2.13e+01	2.12e+01	2.12e+01	3.97e-02	2.10e+01	2.12e+01	2.11e+01	2.11e+01	4.21e-02
F62014	2014	50	5.15e+01	7.32e+01	6.25e+01	6.18e+01	4.54e+00	4.65e+01	6.71e+01	5.76e+01	5.71e+01	4.31e+00	4.47e+01	6.34e+01	5.21e+01	5.25e+01	4.42e+00
F72014	2014	50	9.61e+01	3.63e+02	2.04e+02	2.15e+02	6.34e+01	2.86e+00	4.25e+00	3.46e+00	3.50e+00	2.98e-01	2.50e+00	3.29e+00	2.83e+00	2.83e+00	1.62e-01
F82014	2014	50	3.84e+02	5.98e+02	5.06e+02	5.03e+02	4.44e+01	3.19e+02	5.53e+02	4.40e+02	4.41e+02	4.31e+01	2.96e+02	5.29e+02	4.10e+02	4.06e+02	4.10e+01
F92014	2014	50	4.81e+02	7.77e+02	6.26e+02	6.25e+02	7.26e+01	4.43e+02	7.02e+02	5.70e+02	5.70e+02	6.58e+01	3.97e+02	6.66e+02	5.35e+02	5.35e+02	6.24e+01
F102014	2014	50	8.31e+03	1.34e+04	1.09e+04	1.10e+04	1.07e+03	7.11e+03	1.16e+04	9.77e+03	9.67e+03	1.10e+03	6.43e+03	1.14e+04	8.80e+03	8.89e+03	1.16e+03
F112014	2014	50	1.00e+04	1.48e+04	1.23e+04	1.24e+04	9.81e+02	8.18e+03	1.30e+04	1.09e+04	1.09e+04	1.14e+03	7.66e+03	1.21e+04	9.72e+03	9.80e+03	1.07e+03
F122014	2014	50	3.34e+00	5.64e+00	4.58e+00	4.56e+00	5.21e-01	2.90e+00	4.46e+00	3.60e+00	3.62e+00	3.28e-01	2.12e+00	3.59e+00	2.98e+00	2.93e+00	3.02e-01
F132014	2014	50	5.45e-01	3.51e+00	2.13e+00	1.91e+00	9.38e-01	4.00e-01	8.70e-01	5.98e-01	6.00e-01	1.16e-01	3.54e-01	7.38e-01	5.49e-01	5.43e-01	9.63e-02
F142014	2014	50	2.24e+01	9.04e+01	4.86e+01	4.95e+01	1.41e+01	1.92e-01	8.00e-01	3.43e-01	3.55e-01	7.99e-02	1.92e-01	6.76e-01	2.90e-01	2.96e-01	6.69e-02
F152014	2014	50	2.50e+04	9.64e+05	1.86e+05	2.35e+05	1.92e+05	4.01e+01	5.83e+01	4.77e+01	4.80e+01	4.42e+00	3.22e+01	4.64e+01	4.07e+01	4.01e+01	2.98e+00
F162014	2014	50	2.19e+01	2.37e+01	2.30e+01	2.30e+01	3.52e-01	2.12e+01	2.32e+01	2.26e+01	2.24e+01	4.31e-01	2.06e+01	2.27e+01	2.18e+01	2.18e+01	3.97e-01
F172014	2014	50	2.41e+06	1.36e+08	3.08e+07	3.71e+07	2.52e+07	5.15e+05	3.91e+06	1.71e+06	1.70e+06	6.62e+05	2.65e+05	1.82e+06	7.15e+05	7.61e+05	3.19e+05
F182014	2014	50	5.75e+06	2.19e+07	1.40e+07	1.39e+07	3.64e+06	3.39e+06	1.11e+07	7.01e+06	7.04e+06	1.45e+06	3.32e+06	7.53e+06	4.91e+06	4.91e+06	8.46e+05
F192014	2014	50	1.16e+04	2.25e+08	2.01e+06	1.78e+07	3.90e+07	9.88e+02	1.64e+04	4.99e+03	6.35e+03	3.61e+03	9.88e+02	9.74e+03	1.99e+03	3.48e+03	2.65e+03
F202014	2014	50	3.65e+05	1.01e+11	3.04e+09	8.14e+09	1.61e+10	1.12e+05	3.30e+05	2.29e+05	2.26e+05	4.34e+04	9.33e+04	2.92e+05	2.09e+05	2.09e+05	3.93e+04
F212014	2014	50	1.43e+07	5.95e+08	7.07e+07	1.11e+08	1.10e+08	5.26e+05	2.05e+06	1.09e+06	1.19e+06	4.21e+05	2.92e+05	1.50e+06	6.88e+05	6.98e+05	2.59e+05
F222014	2014	50	3.53e+03	8.83e+06	6.16e+03	1.97e+05	1.23e+06	1.92e+03	6.99e+03	4.68e+03	4.68e+03	1.17e+03	1.10e+03	3.83e+03	2.27e+03	2.28e+03	6.73e+02
F232014	2014	50	2.76e+02	3.46e+02	2.95e+02	3.00e+02	1.72e+01	2.37e+02	2.45e+02	2.39e+02	2.40e+02	1.90e+00	2.36e+02	2.39e+02	2.37e+02	2.37e+02	4.84e-01
F242014	2014	50	3.91e+02	5.43e+02	4.50e+02	4.48e+02	3.36e+01	3.12e+02	3.98e+02	3.36e+02	3.38e+02	1.57e+01	3.03e+02	3.59e+02	3.25e+02	3.27e+02	1.26e+01
F252014	2014	50	2.47e+02	3.51e+02	2.86e+02	2.90e+02	2.31e+01	2.18e+02	3.36e+02	2.58e+02	2.60e+02	2.34e+01	2.09e+02	3.29e+02	2.28e+02	2.38e+02	2.86e+01
F262014	2014	50	1.06e+02	4.75e+02	2.09e+02	2.23e+02	7.85e+01	1.00e+02	4.67e+02	2.02e+02	2.14e+02	7.49e+01	1.00e+02	4.53e+02	2.01e+02	2.12e+02	7.24e+01
F272014	2014	50	8.45e+02	2.43e+03	2.18e+03	2.15e+03	2.18e+02	4.43e+02	2.28e+03	1.93e+03	1.92e+03	2.53e+02	4.21e+02	2.15e+03	1.81e+03	1.79e+03	2.42e+02
F282014	2014	50	5.62e+03	1.17e+04	8.64e+03	8.63e+03	1.43e+03	4.98e+03	1.06e+04	7.81e+03	7.83e+03	1.37e+03	4.71e+03	9.93e+03	7.22e+03	7.15e+03	1.41e+03
F292014	2014	50	4.48e+07	1.82e+09	6.34e+08	7.46e+08	4.75e+08	9.87e+05	6.09e+07	2.15e+06	3.58e+06	8.25e+06	5.50e+05	4.01e+07	1.10e+06	2.02e+06	5.46e+06
F302014	2014	50	1.56e+08	4.59e+09	1.30e+09	1.43e+09	8.31e+08	1.04e+06	4.33e+06	2.29e+06	2.41e+06	7.99e+05	3.94e+05	2.13e+06	7.15e+05	8.50e+05	4.03e+05
F12017	2017	50	5.39e+09	2.29e+10	1.49e+10	1.49e+10	4.07e+09	1.84e+08	3.16e+08	2.38e+08	2.42e+08	2.86e+07	1.32e+08	2.06e+08	1.77e+08	1.75e+08	1.67e+07
F22017	2017	50	5.73e+04	1.46e+05	9.32e+04	9.57e+04	2.04e+04	2.32e+02	4.35e+02	3.16e+02	3.16e+02	3.57e+01	1.67e+02	2.64e+02	2.14e+02	2.14e+02	2.05e+01
F32017	2017	50	2.22e+03	1.47e+04	5.93e+03	6.48e+03	2.27e+03	1.03e+02	2.88e+02	2.05e+02	2.05e+02	4.29e+01	4.70e+01	2.34e+02	1.68e+02	1.69e+02	4.08e+01
F42017	2017	50	8.19e+03	2.71e+04	1.66e+04	1.72e+04	5.17e+03	6.54e+02	9.39e+02	7.65e+02	7.83e+02	7.11e+01	5.68e+02	7.77e+02	6.72e+02	6.74e+02	4.11e+01
F52017	2017	50	2.92e-03	3.51e-02	1.08e-02	1.08e-02	5.74e-03	1.05e-03	1.12e-02	4.22e-03	4.45e-03	2.01e-03	7.58e-04	6.88e-03	2.44e-03	2.60e-03	1.30e-03
F62017	2017	50	3.30e+05	1.47e+06	8.68e+05	9.40e+05	2.55e+05	5.86e+03	1.15e+04	1.01e+04	9.91e+03	9.79e+02	5.33e+03	8.66e+03	7.19e+03	7.12e+03	7.29e+02
F72017	2017	50	8.24e+02	1.36e+03	1.14e+03	1.14e+03	1.37e+02	7.43e+02	1.35e+03	1.01e+03	1.01e+03	1.39e+02	6.35e+02	1.22e+03	9.07e+02	9.10e+02	1.30e+02
F82017	2017	50	1.49e+01	7.81e+01	3.60e+01	4.06e+01	1.24e+01	1.36e+01	5.33e+01	3.01e+01	3.27e+01	8.81e+00	1.32e+01	5.08e+01	2.90e+01	3.12e+01	8.11e+00
F92017	2017	50	1.10e+04	1.48e+04	1.26e+04	1.27e+04	9.77e+02	9.71e+03	1.37e+04	1.13e+04	1.14e+04	9.51e+02	8.46e+03	1.27e+04	1.02e+04	1.01e+04	9.50e+02
F102017	2017	50	4.58e+05	2.00e+08	3.47e+07	5.28e+07	5.33e+07	1.08e+05	3.04e+05	2.01e+05	1.98e+05	4.39e+04	8.59e+04	2.90e+05	1.81e+05	1.82e+05	4.46e+04
F112017	2017	50	8.06e+07	1.51e+09	3.33e+08	4.13e+08	2.64e+08	4.69e+07	1.55e+08	8.69e+07	9.12e+07	2.88e+07	2.48e+07	1.10e+08	5.17e+07	5.58e+07	1.85e+07
F122017	2017	50	3.97e+08	2.48e+10	8.83e+09	9.34e+09	5.59e+09	2.26e+07	6.67e+07	3.56e+07	3.68e+07	8.91e+06	1.20e+07	2.83e+07	2.01e+07	2.04e+07	3.68e+06
F132017	2017	50	7.87e+05	4.17e+07	4.04e+06	5.74e+06	6.46e+06	3.87e+05	1.84e+06	6.38e+05	7.13e+05	2.87e+05	2.28e+05	7.54e+05	3.81e+05	4.02e+05	1.18e+05
F142017	2017	50	6.10e+07	3.06e+09	6.77e+08	9.07e+08	7.04e+08	2.58e+06	1.46e+07	7.01e+06	7.72e+06	2.77e+06	1.72e+06	7.29e+06	3.62e+06	3.77e+06	1.19e+06
F152017	2017	50	1.44e+05	9.05e+08	2.30e+07	1.02e+08	1.68e+08	2.97e+04	1.84e+05	1.15e+05	1.14e+05	3.52e+04	2.06e+04	1.60e+05	9.39e+04	9.55e+04	3.17e+04
F162017	2017	50	3.43e+04	1.42e+05	7.89e+04	8.21e+04	2.41e+04	5.31e+03	2.63e+04	1.19e+04	1.24e+04	4.44e+03	6.58e+02	2.54e+03	1.41e+03	1.46e+03	4.64e+02
F172017	2017	50	1.08e+06	4.36e+07	4.39e+06	7.57e+06	7.71e+06	3.49e+05	3.49e+06	1.33e+06	1.53e+06	8.83e+05	2.86e+05	1.49e+06	6.42e+05	6.91e+05	2.88e+05
F182017	2017	50	9.82e+09	1.67e+11	3.87e+10	4.70e+10	3.34e+10	4.87e+06	1.62e+07	9.81e+06	1.02e+07	2.38e+06	3.28e+06	8.91e+06	6.11e+06	6.05e+06	1.30e+06
F192017	2017	50	2.33e+03	8.78e+03	5.33e+03	5.39e+03	1.46e+03	6.85e+02	4.65e+03	2.45e+03	2.46e+03	1.04e+03	1.84e+02	3.14e+03	9.85e+02	1.08e+03	6.95e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	2.79e+03	2.90e+04	1.11e+04	1.07e+04	5.70e+03	4.83e+02	1.31e+03	1.02e+03	9.03e+02	2.94e+02	4.18e+02	1.07e+03	8.83e+02	7.63e+02	2.32e+02
F212017	2017	50	5.19e+02	1.24e+04	8.12e+02	3.62e+03	4.57e+03	4.12e+02	1.09e+04	6.44e+02	3.02e+03	4.01e+03	3.56e+02	1.03e+04	5.54e+02	2.42e+03	3.53e+03
F222017	2017	50	1.07e+04	5.99e+04	3.35e+04	3.29e+04	1.31e+04	1.09e+03	2.86e+03	2.54e+03	2.06e+03	7.27e+02	9.81e+02	2.46e+03	2.27e+03	1.81e+03	6.16e+02
F232017	2017	50	5.54e+03	3.67e+04	1.37e+04	1.52e+04	6.52e+03	9.34e+02	1.85e+03	1.68e+03	1.52e+03	3.15e+02	8.43e+02	1.63e+03	1.48e+03	1.35e+03	2.70e+02
F242017	2017	50	3.55e+03	9.36e+03	6.39e+03	6.45e+03	1.34e+03	8.09e+02	1.49e+03	9.88e+02	1.01e+03	1.66e+02	7.44e+02	1.11e+03	8.57e+02	8.77e+02	9.19e+01
F252017	2017	50	5.96e+03	3.09e+04	1.31e+04	1.39e+04	4.39e+03	1.09e+03	2.14e+03	1.40e+03	1.47e+03	2.40e+02	2.74e+00	1.98e+03	1.24e+03	1.33e+03	2.90e+02
F262017	2017	50	1.32e+03	2.97e+03	1.97e+03	2.02e+03	4.12e+02	1.03e+03	2.25e+03	1.42e+03	1.47e+03	3.02e+02	8.74e+02	1.83e+03	1.25e+03	1.26e+03	2.35e+02
F272017	2017	50	8.06e+02	2.57e+03	1.43e+03	1.50e+03	4.37e+02	8.75e+01	4.79e+02	1.04e+02	2.33e+02	1.74e+02	8.01e+01	4.53e+02	8.91e+01	1.31e+02	1.17e+02
F282017	2017	50	1.07e+08	1.96e+10	2.47e+09	4.32e+09	4.15e+09	1.16e+05	5.84e+05	2.08e+05	2.17e+05	7.20e+04	9.78e+04	2.39e+05	1.58e+05	1.55e+05	3.21e+04
F292017	2017	50	4.91e+09	6.51e+10	1.64e+10	2.15e+10	1.52e+10	1.02e+07	1.23e+08	2.80e+07	3.16e+07	1.85e+07	5.88e+06	3.41e+07	1.39e+07	1.42e+07	5.53e+06
F12022	2022	20	6.11e+02	1.96e+04	6.17e+03	6.77e+03	4.03e+03	1.83e+01	3.29e+01	2.45e+01	2.46e+01	3.50e+00	1.13e+01	2.18e+01	1.68e+01	1.70e+01	2.32e+00
F22022	2022	20	8.13e+01	2.83e+02	1.46e+02	1.63e+02	5.32e+01	5.35e+00	7.66e+01	5.02e+01	5.70e+01	1.84e+01	4.49e+00	7.62e+01	4.98e+01	5.59e+01	1.81e+01
F32022	2022	20	1.26e-03	1.33e-02	2.29e-03	2.63e-03	1.85e-03	5.05e-04	1.29e-03	1.09e-03	1.06e-03	1.47e-04	4.38e-04	9.67e-04	7.71e-04	7.54e-04	1.17e-04
F42022	2022	20	8.99e+01	3.32e+02	2.07e+02	2.10e+02	5.40e+01	8.96e+01	3.32e+02	1.92e+02	1.98e+02	5.26e+01	8.89e+01	3.18e+02	1.90e+02	1.96e+02	5.11e+01
F52022	2022	20	1.58e+00	1.11e+01	4.73e+00	5.14e+00	2.17e+00	1.05e+00	1.09e+01	4.32e+00	4.83e+00	2.17e+00	1.02e+00	1.07e+01	4.07e+00	4.58e+00	2.22e+00
F62022	2022	20	5.68e+05	3.26e+06	1.93e+06	1.87e+06	6.04e+05	3.60e+05	1.37e+06	8.34e+05	8.43e+05	2.46e+05	1.30e+05	7.26e+05	4.04e+05	3.97e+05	1.31e+05
F72022	2022	20	7.11e+01	8.38e+02	2.59e+02	3.53e+02	2.29e+02	5.68e+01	2.55e+02	9.42e+01	9.96e+01	3.74e+01	4.21e+01	9.27e+01	6.05e+01	6.18e+01	1.17e+01
F82022	2022	20	2.79e+02	4.25e+03	1.87e+03	2.09e+03	1.12e+03	3.36e+01	1.01e+02	4.17e+01	4.41e+01	1.14e+01	2.81e+01	5.22e+01	3.20e+01	3.25e+01	3.37e+00
F92022	2022	20	5.90e+01	1.07e+03	6.64e+02	6.78e+02	1.73e+02	8.40e+00	3.51e+02	3.42e+02	3.29e+02	6.54e+01	5.72e+00	3.39e+02	3.37e+02	3.24e+02	6.49e+01
F102022	2022	20	4.12e+02	4.64e+03	3.23e+03	2.81e+03	1.39e+03	4.08e+02	4.43e+03	2.69e+03	2.18e+03	1.30e+03	4.05e+02	3.78e+03	2.24e+03	1.78e+03	1.20e+03
F112022	2022	20	4.00e+00	2.29e+03	1.59e+02	5.34e+02	6.85e+02	1.28e+00	1.55e+03	1.59e+00	3.19e+01	2.17e+02	1.12e+00	1.55e+00	1.33e+00	1.32e+00	9.06e-02
F122022	2022	20	3.29e+02	8.40e+02	5.49e+02	5.52e+02	1.27e+02	2.91e+02	6.35e+02	4.05e+02	4.20e+02	7.53e+01	2.57e+02	5.25e+02	3.57e+02	3.65e+02	5.94e+01